AMULYA VISWAMBHARAN

Al Ain, UAE - 0525962627 - amulyasumesh@gmail.com - LinkedIn - Google Scholar-UAE Golden Visa

PROFESSIONAL SUMMARY

Ph.D.-qualified Electrical Engineer and dedicated academic with over a decade of teaching and industry experience in renewable energy systems, power electronics, advanced control, and smart grid technologies. Expertise in integrating Artificial Intelligence and Machine Learning (AI/ML) and Digital Twin frameworks for smart grid and optimisation in modern energy applications. Passionate about advancing innovative teaching practices, interdisciplinary research, and developing future-ready engineers in alignment with emerging trends in smart grids and intelligent energy systems.

RESEARCH INTERESTS

Artificial Intelligence in Power Systems | Digital Twin | Smart Grid Optimization | Renewable Energy Integration
Disturbance Observer based Control | Control Systems | Reinforcement Learning

EDUCATION

PhD in Electrical Engineering (GPA: 4.0/4.0) **Aug 2020 - May 2025**
United Arab Emirates University

- Thesis on "Disturbance Observer Based Control for Power Converters in Renewable Energy Systems".

MTech in Power Systems (GPA: 3.6/4.0) **Jul 2012 - May 2014**
National Institute of Technology Calicut

BTech in Electrical & Electronics Engineering (GPA: 3.3/4.0) **Jul 2002 - May 2006**
Cochin University of Science and Technology

ACADEMIC AND RESEARCH EXPERIENCE

Postdoctoral Researcher , UAE University **2026- Present**
Al Ain , UAE

- Conduct research on AI-powered digital twin frameworks for smart grid optimization and resilience
- Develop machine learning-based predictive control, fault detection, and monitoring systems for energy applications
- Design and implement real-time simulation models integrating IoT data for dynamic grid analysis and decision-making
- Publish high-impact research in leading journals and contribute to grant proposals and funded research projects
- Collaborate with multidisciplinary teams and supervise student research in AI-driven energy systems and advanced control technologies

Research Scholar , UAE University **2020 - 2025**
Al Ain , UAE

- Designed and implemented advanced control algorithms for power converters.
- Conducted real-time simulation and hardware testing to validate control strategies.
- Communicate research findings through detailed reports, publications in top 5% IEEE journals, and impactful conference presentations.
- Engage in ongoing learning and skill development to stay current with advancements in control theory, machine learning, and related fields.

Graduate Teaching Assistant UAE University **2020 - 2025**
Al Ain , UAE

- Delivered applied tutorials in Engineering Thermodynamics using Engineering Equation Solver (EES) to support learning by doing.
- Conducted regular assessments to evaluate student progress and ensure the acquisition of key learning objectives.
- Facilitated weekly office hours, providing mentor ship to students and assisting them in reaching their academic milestones.
- Collaborated with faculty members by providing support in various teaching-related tasks.

Assistant Professor, College of Engineering Munnar, Website **2006 - 2017**
Kerala, India

- Delivered practical, student-centred instruction across electrical and electronics engineering courses using projects, simulation, and laboratory-based learning.
- Designed course materials, assessments, and learning activities aligned with program outcomes and academic standards.
- Monitored student progress, provided continuous feedback and academic advising, and supervised student research projects.
- Contributed to curriculum review, examinations, academic administration, faculty teamwork, and departmental activities.

INDUSTRY EXPERIENCE

2017 - 2020

Laboratory Engineer, Geo Chem Middle East **Dubai , UAE**

- Performed testing, evaluation, calibration, and modification of instruments to ensure optimal performance
- Collaborated with multidisciplinary teams to support project execution and technical tasks
- Prepared technical reports and documentation to communicate results effectively
- Ensured compliance with safety standards and maintained up-to-date technical knowledge

TEACHING EXPERIENCE

Courses Taught: Power Systems, Power Electronics, Electrical Machines, Control Systems, Electrical Measurements, Engineering Thermodynamics

Teaching Approach: Learning by Doing; student-centred and project-based learning; outcome-based education (OBE); continuous assessment and feedback; integration of educational technology and simulation tools (MATLAB, EES, PVsyst)

TECHNICAL SKILLS

MATLAB/Simulink, PVsyst, PSCAD/EMTDC, ETAP, Python, LaTeX, EES, AutoCAD, PLC Programming, Open DSS

INVITED TALKS & PROFESSIONAL SERVICE

- Served as a Technical Program Committee (TPC) Member for the 2026 IEEE International Conference on Robotics and Mechatronics (IEEE ICRM 2026)
- Invited Speaker – IEEE IA/IE/PELS Joint Chapter, Kerala Section (GSAC) technical lecture on “AI Powered Digital Twin for Renewable Energy” hosted at NIT Calicut
- Invited resource person for FDP on "Electrical Engineering and Beyond: Research Prospects," Bishop Jerome Institute, Kerala – delivered session on “Energy Digital Twin”
- Reviewer for leading journals and conferences, including IEEE, ScienceDirect (Elsevier), and Springer
- Active participant and presenter at international conferences (IEEE IAS, IECON)

HONORS & DISTINCTIONS

- UAE Golden Visa
- Industry 4.0 Postdoctoral Fellowship
- UAEU PhD Fellowship
- Distinguished Student Award ,UAEU

TRAINING AND CERTIFICATIONS

- OpenDSS Skill Development Workshop, IEEE PES Kerala Chapter & IEEE YP PES
- “AI in Electrical Tech: Enhancing Design,Control, & Maintenance”Tech Source Systs &Ascendas Systs
- “Predictive Maintenance with Digital Twins and Machine Learning”, NEXUS OFS
- ”AI for Smart Cities”, student tutorial by BIDAC, UAEU
- ”Machine Learning for All”, online course authorized by the University of London
- ”Daily Study Group: Introduction to Machine Learning”, program by Wolfram U
- ”Introduction to Artificial Intelligence”, program by Simplilearn SkillUP
- PhD students Teaching Academy Program Fifth Cohort, United Arab Emirates University
- “SCADA and its Applications”, Kerala State Electricity Board
- Coordinated a workshop on “Power System Modeling and Simulation”, NIT Calicut
- FDP on “Advanced Control: Theory and Applications”, NIT Calicut
- FDP on “Research and Technical challenges of Smart Grid”, NIT Calicut
- FDP on “Smart Grid Engineering”, NIT Calicut

PUBLICATIONS: h-index: 4 Citations: 110 Published : 16

Journal

1) **A. Viswambharan**, M. Ali, W. Eldosh, B. Shawabkeh, AF. Hasan , and R. Errouissi, “Design, simulation, and multi scenario optimization of a grid connected photovoltaic system using PVsyst: A case study at UAEU. Renewable Energy Focus. 2026 Aug 1:100912.

2) **A. Viswambharan**, R. Errouissi, M. Debouza, H. Shareef, and A. Wahyudie, “Disturbance Observer-Based Control for PMSG Wind Turbine System to Cope With Both Balanced and Unbalanced Grid Voltages” IEEE Transactions on Industry Applications, 2026, doi: 10.1109/TIA.2026.3680377.

3) **A. Viswambharan**, R. Errouissi, M. Debouza, H. Shareef, and A. Wahyudie, “Disturbance Observer-Based Robust Current Control Scheme for Three-Phase Grid-Tied Inverters With LCL Filter Considering Unbalanced Grid Voltages,” IEEE Transactions on Power Electronics, pp. 1–19, 2024, doi: 10.1109/TPEL.2024.3478336.

4) **A. Viswambharan**, R. Errouissi, Debouza and H. Shareef "Experimental Verification of Disturbance Observer-Based Backstepping Control for DC–DC Boost Converter." IEEE Transactions on Circuits and Systems I: Regular Papers (2023). Doi:10.1109/TCSI.2023.3318970

5) R. Errouissi, **A. Viswambharan**, ,and H. Shareef “Robust Full Linear State-Feedback Controller for Grid-Tied LCL Filter Systems Using Adaptive High-Gain Observer,” IEEE Transactions on Industry Applications, March. 2023,doi:10.1109/TIA.2023.3262500

6) R. Errouissi, H. Shareef, **A. Viswambharan**, and A. Wahyudie, “Disturbance-Observer-Based Feedback Linearization Control for Stabilization and Accurate Voltage Tracking of a DC–DC Boost Converter,” IEEE Transactions on Industry Applications, vol. 58, no. 5, pp. 6687–6700, Sep. 2022, doi: 10.1109/TIA.2022.3183040.

7) **A. Viswambharan**, “Comparison of SCIG and DFIG Wind Turbines during Variable Wind Speed,” International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering, vol. 4, no. 2, pp. 1–4, Feb. 2016, doi:10.17148/IJIREEICE.2016.4201.

Conference

1) Husain MS, Fatima K, Hussain MS, **A. Viswambharan** , Idrisi AH. Machine Learning Based Water Quality Classification in Geothermally Active Regions. In2025 Advances in Science and Engineering Technology International Conferences (ASET) 2025 Sep 15 (pp. 1-6). IEEE.

2) Fatima K, Shareef H, Hussain MS, Husain MS, **A. Viswambharan** , Idrisi AH. “Hurricane-Induced Line Failure Risk Classification Using Machine Learning”,in IEEE 19th International Conference on Application of Information and Communication Technologies (AICT) Oct 2025

3) **A. Viswambharan** , Rachid Errouissi, KSP Kiranmai, and Mahdi Debouza “ Artificial Neural Network Controller for DC-DC Boost Converter: A Design and Performance Analysis”, in SPIES”24, December 2024, Abu Dhabi, UAE

4) **A. Viswambharan** and Rachid Errouissi, “Disturbance Observer Based Control for Torque Ripple Mitigation in PMSG-based Wind Turbine During Unbalanced Grid Voltages”, in 2023 IEEE Industry Applications Society Annual Meeting IAS), Oct.29-Nov 2, 2023, Nashville, USA

5) Rachid Errouissi, **A. Viswambharan** , and Hussain Shareef, “Robust Full Linear State-Feedback Controller for Grid-Tied LCL Filter Systems Using Adaptive High-Gain Observer,” in 2022 IEEE Industry Applications Society Annual Meeting IAS), Oct.9-13, 2022, Detroit, Michigan , USA

REFERENCE

Dr. Rachid Errouissi, Associate Professor, Electrical & Communication Eng., UAEU, rachid.errouissi@uaeu.ac.ae